

Is a parabolic flow profile a defensible idealisation for the vessels in this study?

Working note for the OCTA vessel-diameter manuscript

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Why this note exists

Section 4.7 proposes a forward model that starts from a vessel of circular cross-section carrying a parabolic flow profile. This note checks that assumption against the vessels actually measured here, so that the claim rests on arithmetic rather than on a rule of thumb. Everything needed to reproduce the numbers is below.

Inputs

Quantity	Value used	Where it comes from
Vessel diameter d	26 μm to 62 μm	ground truth, $n = 7$, this study (flicker set: 28 μm to 48 μm , $n = 5$)
Blood density ρ	1060 kg m^{-3}	standard value for whole blood, uncited
Blood viscosity μ	3.5 mPa s, swept 2.5 mPa s to 4.0 mPa s	bulk value; see the note on apparent viscosity [1]
Mean velocity v	derived, see below	total arterial flow 2.82 $\mu\text{L min}^{-1}$ over six vessels [2]
Heart rate f	5 Hz to 10 Hz (300 to 600 bpm)	swept; anaesthesia lowers heart rate and cardiac output [4]

Velocity. No measured velocity exists for these vessels, so it is derived. Zhi *et al.* measured a total arterial flow of 2.82 $\mu\text{L min}^{-1}$ in the mouse retina across the six main vessels [2]. Per vessel that is $Q = 2.82/6/60 = 7.83 \times 10^{-3} \text{ mm}^3 \text{ s}^{-1}$, and the mean velocity follows from continuity,

$$v = \frac{Q}{\pi(d/2)^2}. \quad (1)$$

This is an estimate from a different mouse strain under a different anaesthetic, and their wording leaves it open whether those six are six arteries or six vessels in total. If the per-vessel flow were twice what is assumed here, every Reynolds number below would double, reaching at most 0.23. The conclusion is therefore also stress-tested at a velocity far above anything plausible here.

Apparent viscosity. In tubes of this calibre the apparent viscosity of blood is lower than the bulk value, the Fåhræus–Lindqvist effect, quantified over diameters from 3.3 μm to 1978 μm by Pries *et al.* [1]. A lower viscosity raises the Reynolds number, so the sweep down to 2.5 mPa s is the conservative direction.

The three numbers

$$\text{Reynolds number} \quad \text{Re} = \frac{\rho v d}{\mu} \quad (2)$$

$$\text{Entrance length} \quad L_e = [0.619 \text{Re}^{1.6} + (0.0567 \text{Re})^{1.6}]^{1/1.6} d \quad (3)$$

$$\text{Womersley number} \quad \alpha = \frac{d}{2} \sqrt{\frac{2\pi f \rho}{\mu}} \quad (4)$$

Re decides laminar against turbulent. L_e decides whether the parabola has room to form before the measurement site. α decides whether pulsatility blunts the profile: for $\alpha \ll 1$ the flow is quasi-steady and stays parabolic through the cardiac cycle.

A caution on L_e . The textbook correlation $L_e \approx 0.06 \text{Re} d$ is not usable at these Reynolds numbers. It is fitted at moderate Re and tends to zero as $\text{Re} \rightarrow 0$, which is unphysical: in creeping flow the development length tends to a constant of roughly $0.6 d$. The expression above carries that limit, and it is what the table uses; it is Eq. (16) of Durst *et al.*, whose constants are the $\text{Re} \rightarrow 0$ and $\text{Re} \rightarrow \infty$ limits of L/D [3].

Results

d (μm)	v (mm s^{-1})	Re	Re, μ 2.5 mPas to 4.0 mPas	L_e (μm)	α (5 Hz)	α (10 Hz)
26	14.8	0.116	0.102–0.163	16	0.040	0.057
30	11.1	0.101	0.088–0.141	19	0.046	0.065
38	6.9	0.079	0.070–0.111	24	0.059	0.083
48	4.3	0.063	0.055–0.088	30	0.074	0.105
62	2.6	0.049	0.043–0.068	38	0.096	0.135

Stress test. Forcing 20 mm s^{-1} through the largest vessel, several times the derived velocity, still gives $\text{Re} = 0.38$ and an entrance length of $38 \mu\text{m}$, that is $0.63 d$.

What this supports, and what it does not

The flow in these vessels is laminar by roughly four orders of magnitude, fully developed within about two thirds of one diameter, which is short against the spacing of bifurcations, and quasi-steady. A parabolic profile is therefore a defensible starting point for a forward model of the OCTA signal.

Three things it does not establish. First, the profile of the *OCTA signal* is not the profile of the flow: the signal also depends on the RBC orientation artefact, on fringe washout, and on the nonlinear relation between signal and flow speed, which is why those terms appear in the proposed model rather than flow alone. Second, the velocity is inferred, not measured; the DyC-OCT data behind this manuscript could supply a measured velocity for exactly these vessels and would remove that assumption. Third, a parabolic profile is the profile of the *plasma and cell mixture* treated as a continuum; it says nothing about the cell-free layer at the wall, which is the very feature that separates the OCTA-visible red blood cell column from the lumen boundary seen with contrast.

References

- [1] A. R. Pries, D. Neuhaus, and P. Gaehtgens, “Blood viscosity in tube flow: dependence on diameter and hematocrit,” *Am. J. Physiol.* **263**(6 Pt 2), H1770–H1778 (1992). [10.1152/ajp-heart.1992.263.6.H1770](#)
- [2] Z. Zhi, X. Yin, S. Dziennis, T. Wietecha, K. L. Hudkins, C. E. Alpers, and R. K. Wang, “Optical microangiography of retina and choroid and measurement of total retinal blood flow in mice,” *Biomed. Opt. Express* **3**(11), 2976–2986 (2012). [10.1364/BOE.3.002976](#)
- [3] F. Durst, S. Ray, B. Ünsal, and O. A. Bayoumi, “The development lengths of laminar pipe and channel flows,” *J. Fluids Eng.* **127**(6), 1154–1160 (2005). Source of the L_e expression, their Eq. (16) for pipe flow.
- [4] B. J. A. Janssen, T. De Celle, J. J. M. Debets, A. E. Brouns, M. F. Callahan, and T. L. Smith, “Effects of anesthetics on systemic hemodynamics in mice,” *Am. J. Physiol. Heart Circ. Physiol.* **287**(4), H1618–H1624 (2004). Cited for the direction of the effect only; the abstract reports cardiac output and blood pressure, not heart rate. [10.1152/ajpheart.01192.2003](#)